(iv) $2^{\left(-\frac{2}{3}\right)}\left[1-\frac{x}{3}+\frac{5}{36}x^2-\frac{5}{81}x^3+\ldots\right]$; $|x|<2$ (2) $(1001)^{\left(\frac{1}{3}\right)} \approx 10.00333$

(5) (i) $1+5x+\frac{25x^2}{2}+\frac{125x^3}{6}+\frac{625x^4}{24}+\frac{625x^5}{24}+\ldots$

(ii) $1-2x+2x^2-\frac{4x^3}{3}+\frac{2x^4}{3}-\frac{4x^5}{15}+\ldots$ (iii) $1+\frac{1}{2}x+\frac{1}{8}x^2+\frac{1}{48}x^3+\frac{1}{384}x^4+\ldots$

(6)(i) $4x-8x^2+\frac{64x^3}{3}-64x^4+\ldots$; $|x|<\frac{1}{4}$

(ii) $-2x-\frac{4x^2}{2}-\frac{8x^3}{3}-\frac{16x^4}{4}-\ldots$; $|x|<\frac{1}{2}$

(iii) $2[3x+\frac{27x^3}{3}+\frac{243x^5}{5}+\frac{2187x^7}{7}+\ldots]$; $|x|<\frac{1}{3}$

(iv) $-2[2x+\frac{8x^3}{3}+\frac{32x^5}{5}+\frac{128x^7}{7}+\ldots]$; $|x|<\frac{1}{2}$

(8) $\left(\frac{15}{16}\right)^{\frac{1}{8}} \simeq 0.99196$ (9) $\frac{28}{3}$ (10) $\frac{1}{2}log_e 10$

பயிற்சி 5.5

1	2	3	4	5	6	7	8	9	10
(4)	(4)	(4)	(4)	(2)	(3)	(1)	(3)	(4)	(1)
11	12	13	14	15	16	17	18	19	20
(1)	(4)	(4)	(2)	(3)	(2)	(2)	(3)	(3)	(2)

பயிற்சி 6.1

(1) (i) $x^2+y^2=81$ (ii) $\frac{x^2}{81}+\frac{y^2}{36}=1$ (2) (i) $y=\pm 2$ (ii) $x=\pm 3$

(3) $x^{2/3}+y^{2/3}=a^{2/3}$ (4) $k=-24,\ b=-\frac{1}{4}$ (5) $x^2+y^2=16$

(6) $x^2+y^2-4x-4y+8=0$ (7) $x^2+y^2-5x+8y+16=0$

(8) $y^2=2x$ (10) $8x^2+36y^2-16x+252y+431=0$

(11) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (12) $9x^2-33x-3y+35=0$

(13) $x^2+y^2-12x+9y+63=0$, கற்பனை வட்டம்

(14) $(5-\sqrt{21},3),(5+\sqrt{21},3),(8,-3),(2,-3)$

(15) $\frac{x^2}{25}+\frac{y^2}{9}=1$

பயிற்சி 6.2

(1)(i) $y=5x-4$ (ii) $3x-y=2$ (iii) $2x+3y=5$ (iv) $x+\sqrt{3}y=(1+\sqrt{3})$

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